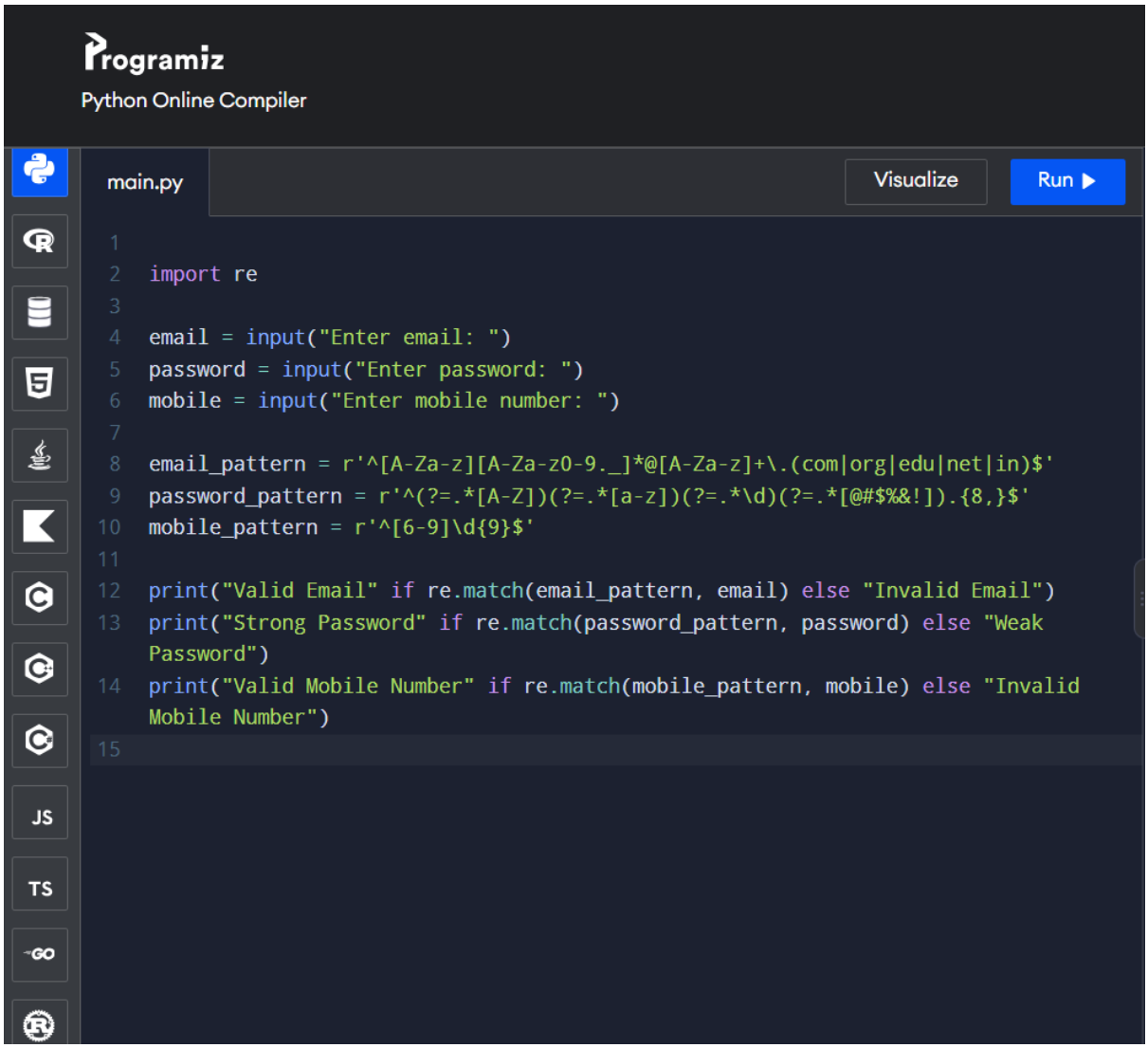
NAME: VIJAYALAKSHMI S

REG NO:192472326

Section 1: Intelligent Email and Password Validator using Regular Expressions



Output

```
Enter email: student123@gmail.com
Enter password: Hello@123
Enter mobile number: 9876543210
Valid Email
Strong Password
Valid Mobile Number
```

Section 2: Design a Finite State Automata (FSA) Simulator

Programiz
Python Online Compiler

main.py | Visualize | Run ▶

```
    'q0': {'a': 'q1', 'b': 'q0'},
    'q1': {'a': 'q1', 'b': 'q2'},
    'q2': {'a': 'q1', 'b': 'q0'}
}

n = int(input("Number of strings: "))

for i in range(n):
    s = input("Enter string: ")
    state = start
    path = [state]

    for ch in s:
        if ch not in alphabet:
            state = None
            break
        state = trans[state][ch]
        path.append(state)

    print("Transition Path:", " -> ".join(path))

    if state in final:
        print("Accepted")
    else:
        print("Rejected")
```

Output

```
Number of strings: 2
Enter string: abaab
Transition Path: q0 -> q1 -> q2 -> q1 -> q1 -> q2
Accepted
Enter string: aabb
Transition Path: q0 -> q1 -> q1 -> q2 -> q0
Rejected
```

Section 3: Smart Pattern Matching Engine using Regular Expressions

main.py

Visualize

Run

```
import re

text = """Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing"""

print("1.Search Date")
print("2.Search Phone Number")
print("3.Search Hashtag")
print("4.Search Mention")
print("5.Search Prefix")
print("6.Search Suffix")
print("7.Search Word")

choice = input("Enter choice: ")

patterns = {
    '1': r'\b\d{2}/\d{2}/\d{4}\b',
    '2': r'\b[6-9]\d{9}\b',
    '3': r'#[A-Za-z0-9_]+',
    '4': r'@[A-Za-z0-9_]+'
}
```

```
    '4': r'@[A-Za-z0-9_]+'
}

if choice in patterns:
    result = re.findall(patterns[choice], text)
elif choice == '5':
    word = input("Enter prefix: ")
    result = re.findall(r'\b' + re.escape(word) + r'\w*', text, re.I)
elif choice == '6':
    word = input("Enter suffix: ")
    result = re.findall(r'\b\w*' + re.escape(word) + r'\b', text, re.I)
elif choice == '7':
    word = input("Enter word: ")
    result = re.findall(r'\b' + re.escape(word) + r'\b', text, re.I)
else:
    result = []

print("Matching patterns:", result)
```

Output

```
1.Search Date
2.Search Phone Number
3.Search Hashtag
4.Search Mention
5.Search Prefix
6.Search Suffix
7.Search Word
Enter choice: 1
Matching patterns: ['12/09/2026']
```